Week_3_Rupa_Shrestha

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```r
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#Week3 Assignment
```

##Read the data and display the first and last six rows of the data.

```r
library(readxl)

housing_data=read_excel("D:/Webster University/Assignments/Analytics
Programming/Week-3/house_selling_prices_FL.xlsx")

head(housing_data)

## # A tibble: 6 × 9
##   House Taxes Bedrooms Baths Quadrant    NW  price  size   lot
##   <dbl> <dbl>    <dbl> <dbl> <chr>    <dbl>  <dbl> <dbl> <dbl>
## 1     1  1360        3   2   NW           1 145000  1240 18000
## 2     2  1050        1   1   NW           1  68000   370 25000
## 3     3  1010        3   1.5 NW           1 115000  1130 25000
## 4     4   830        3   2   SW           0  69000  1120 17000
## 5     5  2150        3   2   NW           1 163000  1710 14000
## 6     6  1230        3   2   NW           1  69900  1010  8000

tail(housing_data)

## # A tibble: 6 × 9
##   House Taxes Bedrooms Baths Quadrant    NW  price  size   lot
##   <dbl> <dbl>    <dbl> <dbl> <chr>    <dbl>  <dbl> <dbl> <dbl>
## 1    95  2880        4     2 NW           1 147000  1860 35000
## 2    96   990        2     2 NW           1 176000  1060 27500
## 3    97  3030        3     2 SW           0 196500  1730 47400
## 4    98  1580        3     2 NW           1 132200  1370 18000
## 5    99  1770        3     2 NE           0  88400  1560 12000
## 6   100  1430        3     2 NW           1 127200  1340 18000
```

##Rename ALL columns with FOUR letter words.

```r
library(stringr)

rename_four_letters_columns <- function(df) {
  column_names <- colnames(df)
  new_names <- sapply(column_names, function(name) {
    if (nchar(name) > 4) {
      paste0(str_sub(name, 1, 4))
    } else if (nchar(name) < 4) {
```

```r
      paste0(str_pad(name, 4, side = "right", pad = "1"))
    } else {
      name
    }
  })
  colnames(df) <- new_names
  return(df)
}

df <- rename_four_letters_columns(housing_data)

colnames(df)

## [1] "Hous" "Taxe" "Bedr" "Bath" "Quad" "NW11" "pric" "size" "lot1"
```

Use the str() function to view the column names and data types for the table.

```r
str(df)

## tibble [100 × 9] (S3: tbl_df/tbl/data.frame)
##  $ Hous: num [1:100] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Taxe: num [1:100] 1360 1050 1010 830 2150 1230 150 1470 1850 320 ...
##  $ Bedr: num [1:100] 3 1 3 3 3 3 2 3 3 3 ...
##  $ Bath: num [1:100] 2 1 1.5 2 2 2 2 2 2 2 ...
##  $ Quad: chr [1:100] "NW" "NW" "NW" "SW" ...
##  $ NW11: num [1:100] 1 1 1 0 1 1 1 1 1 1 ...
##  $ pric: num [1:100] 145000 68000 115000 69000 163000 ...
##  $ size: num [1:100] 1240 370 1130 1120 1710 1010 860 1420 1270 1160 ...
##  $ lot1: num [1:100] 18000 25000 25000 17000 14000 8000 15300 18000 16000
8000 ...
```

##Check the 'type of ' variables Bedrooms and Baths. If they are not character, convert them to character

```r
class(housing_data$Bedrooms)

## [1] "numeric"

class(housing_data$Baths)

## [1] "numeric"

housing_data$Bedrooms <- as.character(housing_data$Bedrooms)
housing_data$Baths <- as.character(housing_data$Baths)

class(housing_data$Bedrooms)

## [1] "character"

class(housing_data$Baths)

## [1] "character"
```

##Display the value counts for Quadrant.

```
value_count <- table(housing_data$Quadrant)
print(value_count)

##
## NE NW SE SW
## 17 75  1  7
```

Add a column named PercentTax that contains the percentage of Tax out of the total price of the house.

```
housing_data$PercentTax <- (housing_data$Taxes / housing_data$price) * 100
print(housing_data$PercentTax)

##   [1] 0.93793103 1.54411765 0.87826087 1.20289855 1.31901840 1.75965665
##   [7] 0.30000000 1.07299270 1.52514427 0.45714286 0.97674419 1.06586826
##  [13] 1.42233857 1.48543689 0.92079208 1.18000000 1.23529412 0.08888889
##  [19] 0.96666667 0.99248120 1.49171271 1.07307692 0.47719298 1.15000000
##  [25] 1.53333333 1.90804598 1.36593592 2.21428571 1.39864865 1.00000000
##  [31] 1.28409091 2.03468208 1.52777778 1.12849162 1.44970414 0.90769231
##  [37] 0.76326003 1.28000000 1.97000000 2.06000000 1.98000000 1.03071672
##  [43] 1.18012422 0.86885246 2.05194805 2.51666667 1.14173228 1.12790698
##  [49] 1.31578947 1.48613678 0.93333333 1.01234568 1.09042553 0.83529412
##  [55] 0.93430657 1.93756728 0.02150538 0.73193047 0.92775665 1.68000000
##  [61] 0.25641026 0.41666667 1.54216867 1.93333333 1.10622711 1.60074627
##  [67] 1.64073550 1.38571429 1.00000000 0.87875417 0.88321168 1.50485437
##  [73] 1.53005464 1.82857143 0.86875000 1.46875000 2.19230769 1.81300813
##  [79] 0.09523810 1.77647059 1.01573677 1.23200000 1.09471095 1.86105800
##  [85] 1.61473088 1.12238806 1.10671937 3.11918850 0.95440085 0.59637913
##  [91] 1.23636364 1.54035088 1.53333333 1.68134508 1.95918367 0.56250000
##  [97] 1.54198473 1.19515885 2.00226244 1.12421384

print(housing_data)

## # A tibble: 100 × 10
##    House Taxes Bedrooms Baths Quadrant    NW  price  size   lot PercentTax
##    <dbl> <dbl> <chr>    <chr> <chr>    <dbl>  <dbl> <dbl> <dbl>      <dbl>
##  1     1  1360 3        2     NW           1 145000  1240 18000      0.938
##  2     2  1050 1        1     NW           1  68000   370 25000      1.54
##  3     3  1010 3        1.5   NW           1 115000  1130 25000      0.878
##  4     4   830 3        2     SW           0  69000  1120 17000      1.20
##  5     5  2150 3        2     NW           1 163000  1710 14000      1.32
##  6     6  1230 3        2     NW           1  69900  1010  8000      1.76
##  7     7   150 2        2     NW           1  50000   860 15300      0.3
##  8     8  1470 3        2     NW           1 137000  1420 18000      1.07
##  9     9  1850 3        2     NW           1 121300  1270 16000      1.53
## 10    10   320 3        2     NW           1  70000  1160  8000      0.457
## # ℹ 90 more rows
```

Add a column named PercentSize that contains the percentage of house size out of the total lot size of the house.

```
housing_data$PercentSize = (housing_data$size/housing_data$lot) * 100
print(housing_data$PercentSize)

##   [1]  6.888889  1.480000  4.520000  6.588235 12.214286 12.625000
5.620915
##   [8]  7.888889  7.937500 14.500000 10.166667  5.633333  8.903226
9.464286
##  [15]  6.562500  3.484163 11.750000 30.285714  7.428571  5.000000
3.190661
##  [22]  8.520000  5.318182  7.894737 13.950000  5.885714  6.250000
4.631579
##  [29] 11.071429 12.083333 11.111111  9.642857  6.571429 10.040000
8.184211
##  [36]  5.866667  7.724138 11.100000  5.440000  5.000000  5.000000
8.222222
##  [43]  7.562189  7.769231 10.166667  9.371429  4.272727 13.166667
10.583333
##  [50]  5.951220 12.666667  7.538462 10.648148 11.440000  7.666667
5.611111
##  [57] 44.500000  7.225806  7.600000  8.100000 13.500000 10.384615
14.333333
##  [64] 11.571429  8.333333  7.027778 11.333333 19.166667  8.216216
21.333333
##  [71]  9.000000 12.666667  8.826087  6.915423  8.545455  9.266667
4.962963
##  [78]  4.272727  6.444444 11.750000 25.555556  7.666667  7.313433
7.227273
##  [85]  7.741935  8.727273  7.413793  7.333333  5.548387 10.250000
6.263736
##  [92]  7.361111 13.250000 11.062500  5.314286  3.854545  3.649789
7.611111
##  [99] 13.000000  7.444444

print(housing_data)

## # A tibble: 100 × 11
##     House Taxes Bedrooms Baths Quadrant    NW  price  size   lot PercentTax
##     <dbl> <dbl> <chr>    <chr> <chr>    <dbl>  <dbl> <dbl> <dbl>      <dbl>
##  1      1  1360 3        2     NW           1 145000  1240 18000      0.938
##  2      2  1050 1        1     NW           1  68000   370 25000      1.54
##  3      3  1010 3        1.5   NW           1 115000  1130 25000      0.878
##  4      4   830 3        2     SW           0  69000  1120 17000      1.20
##  5      5  2150 3        2     NW           1 163000  1710 14000      1.32
##  6      6  1230 3        2     NW           1  69900  1010  8000      1.76
##  7      7   150 2        2     NW           1  50000   860 15300      0.3
##  8      8  1470 3        2     NW           1 137000  1420 18000      1.07
##  9      9  1850 3        2     NW           1 121300  1270 16000      1.53
## 10     10   320 3        2     NW           1  70000  1160  8000      0.457
```

```
## # ℹ 90 more rows
## # ℹ 1 more variable: PercentSize <dbl>
```

Use the apply() function to display the total PercentTax and total PercentSize.

```
percent_size_matrix <- as.matrix(housing_data['PercentSize'])
percent_size_sum <- apply(percent_size_matrix,2,sum)

percent_tax_matrix <- as.matrix(housing_data['PercentTax'])
percent_tax_sum <- apply(percent_tax_matrix,2,sum)

print(percent_size_sum)

## PercentSize
##    924.9204

print(percent_tax_sum)

## PercentTax
##   127.9699
```

1. Group and summarize the data that includes newly added columns.

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

summary_data <- housing_data %>%
  group_by(Bedrooms) %>%
  summarize(
    Total_Price = sum(price),
    Average_Percent_Tax=mean(PercentTax),
    Average_Percent_Size=mean(PercentSize),
    .groups="drop"
    )

print(summary_data)

## # A tibble: 5 × 4
##   Bedrooms Total_Price Average_Percent_Tax Average_Percent_Size
##   <chr>          <dbl>               <dbl>                <dbl>
## 1 1              68000                1.54                 1.48
## 2 2            1846300                1.14                 7.36
## 3 3            7682200                1.26                 9.78
```

```
## 4 4          2955000                1.38                9.84
## 5 5           118300                3.12                7.33
```

Filter the data so that it only includes 3-Bedroom houses.

```
filtered_housing_data = housing_data[housing_data$Bedrooms == 3,]
print(filtered_housing_data)
```

```
## # A tibble: 63 × 11
##    House Taxes Bedrooms Baths Quadrant    NW  price  size   lot PercentTax
##    <dbl> <dbl> <chr>    <chr> <chr>    <dbl>  <dbl> <dbl> <dbl>      <dbl>
##  1     1  1360 3        2     NW           1 145000  1240 18000      0.938
##  2     3  1010 3        1.5   NW           1 115000  1130 25000      0.878
##  3     4   830 3        2     SW           0  69000  1120 17000      1.20
##  4     5  2150 3        2     NW           1 163000  1710 14000      1.32
##  5     6  1230 3        2     NW           1  69900  1010  8000      1.76
##  6     8  1470 3        2     NW           1 137000  1420 18000      1.07
##  7     9  1850 3        2     NW           1 121300  1270 16000      1.53
##  8    10   320 3        2     NW           1  70000  1160  8000      0.457
##  9    11   630 3        2     NE           0  64500  1220 12000      0.977
## 10    12  1780 3        2     NW           1 167000  1690 30000      1.07
## # ℹ 53 more rows
## # ℹ 1 more variable: PercentSize <dbl>
```